reachability distance is undefined, and so we still don't know if P1 is an outlier or part of a cluster: we can only tell after the next step, shown in part **(A)** of figure 12.18, by checking the reachability distance (from C1) of the next point P2.

Since this value is smaller than ε, (in the example, 0.8), then we add P2 to C1 and move to the next point P3, also added to C1 as shown in **(B)** and **(C)**. From a visual point of view, in the reachability chart you can see that the reachability distance for P2 is below the threshold line (parallel to the horizontal axis) for ε=0.8.

When we get to P4, we realize that its reachability distance from C1 (or technically from any of the points processed before P4) is larger than ε, and hence P4 is not reachable from C1: since the points are processed in order of reachability distance from current cluster, this means that there is no point still to be processed that can be reachable from points in C1 if the radius of core regions is equal to ε; hence we "close" current cluster, and start a new one.

We start a new cluster for P4, but we don't know yet how to classify it: when we check P5's reachability distance **(D)**, we discover that it is also larger than ε, and this means that the algorithm has detected P4 is an outlier (because it's not reachable by any point in the dataset, given parameters minPoints and ε). Notice that the radius that matters here is ε: by using a higher value for εMAX we have simply instructed OPTICS to filter out as noise only those points with reachability distance larger than εMAX, while for points with reachability distance at most εMAX, we simply defer the decision to this second step: in practice, this allow us to